Motion Under Gravity

Class 11 Physics | Nine Education IIT Academy

Name: ____________________ Date: __________

Take $g = 10\ m/s^2$ unless stated otherwise.

1. A ball is dropped from the top of a 45 m tower. Find the time taken to reach the ground and its speed just before impact.

2. A stone is released from rest at a height of 80 m. Find the time taken to fall and the velocity with which it strikes the ground.

3. A coin is dropped from a hot-air balloon at rest, 125 m above the ground. Find the time it takes to land.

4. A ball dropped from rest falls freely under gravity. Find the distance it covers in the first 2 s, and in the first 4 s.

5. A ball is thrown vertically downward with a speed of 5 m/s from the top of a 30 m tower. Find the time taken to reach the ground.

6. A ball is thrown downward with an initial speed of 10 m/s from a 75 m tower. Find the time it takes to hit the ground.

7. A ball is thrown down from a tower with an initial speed of 5 m/s and reaches the ground in 4 s. Find the height of the tower.

8. A ball dropped from a height strikes the ground with a speed of 60 m/s. Find the height from which it was dropped and the time it took to fall.

9. A ball is dropped from a tower and takes 7 s to reach the ground. Find the height of the tower and its speed just before landing.

10. A ball is thrown vertically downward from the top of a tower with a speed of 4 m/s and hits the ground with a speed of 24 m/s. Find the height of the tower and the time taken.

11. A pebble is dropped into a well and takes 2 s to reach the water surface (ignore the time for sound to travel back). Find the depth of the well.

12. A ball is thrown vertically upward with a speed of 20 m/s. Find the time taken to reach the highest point and the maximum height attained.

13. A stone is projected upward with a velocity of 30 m/s. Find the maximum height reached and the total time of flight, assuming it returns to the point of projection.

14. A ball thrown straight up reaches a maximum height of 20 m. Find its initial speed and the time taken to reach the top.

15. A player throws a ball vertically upward, and it returns to his hands after 8 s. Find the initial speed of the ball and the maximum height reached.

16. A ball is thrown vertically upward with a speed of 10 m/s. Find its velocity and displacement 1.5 s after it is thrown.

17. A stone is thrown vertically upward with an initial speed of 40 m/s. Find its velocity 3 s after projection, and state whether it is still rising or has begun to fall.

18. A ball thrown vertically upward takes 3 s to reach its highest point. Find its initial velocity and the maximum height reached.

19. A ball is thrown upward with a velocity of 30 m/s. Find the two instants at which it is at a height of 25 m above the point of projection.

20. A stone is thrown vertically upward with a speed of 20 m/s. Find its velocity when it is at a height of 15 m — once on the way up and once on the way down.

21. A ball is thrown vertically upward with a speed of 25 m/s. Find its height above the ground and its velocity 2 s after projection.

22. A stone is thrown up with a speed of 35 m/s. Find its velocity 5 s after projection, and state its direction of motion.

23. A ball is thrown vertically upward with a speed of 15 m/s. Show that its displacement after 1 s equals its displacement after 2 s, and explain why using the idea of symmetry about the highest point.

24. A ball dropped from the top of a tower is found to cover 45 m in the last second before hitting the ground. Find the total time of fall and the height of the tower.

25. A body falls freely from rest. Find the ratio of the distances it covers in the 1st, 2nd, and 3rd seconds of its fall.

26. A ball dropped from a certain height covers 20 m in the last second of its fall. Find the total time of fall and the height of the tower.

27. A ball is thrown vertically upward with a speed of 25 m/s. Find the displacement in the 1st second, and the displacement in the 4th second. Explain why the second value is smaller, and state its direction.

28. A ball thrown up at 20 m/s from the ground is at a height of 15 m above the ground at two instants — once going up, once coming down. Find both instants and the time interval between them.

29. A ball is dropped from height H and strikes the ground with speed v after time t. If an identical ball is dropped from height 4H instead, find the new landing speed and new time of fall, in terms of v and t.

30. A ball thrown vertically upward with speed u reaches a maximum height H in time T. If it were thrown instead with speed 2u, find the new maximum height and the new time of flight, in terms of H and T.

31. Two identical balls are dropped from the same height, one released 1 s after the other. If the first ball takes 4 s to reach the ground, find the ratio of the speeds of the two balls at the instant the first one lands.

32. A stone dropped from rest at the top of a tower passes a 2 m tall window in 0.2 s while falling. Find its velocity at the moment it enters the top of the window.

33. A ball is dropped from the top of a 100 m tower at the same instant that another ball is thrown vertically upward from the ground with a velocity of 40 m/s. Find when, and at what height above the ground, the two balls meet.

34. Two balls are dropped from the same point, the second released 2 s after the first. Find the separation between them 5 s after the first ball is dropped, and state how the separation changes with time thereafter.

35. A balloon is rising vertically with a constant velocity of 5 m/s. When it is 30 m above the ground, a stone is released from it. Find the time taken by the stone to reach the ground and its speed on landing.

36. A ball is thrown vertically upward with a speed of 15 m/s from the edge of a cliff 20 m high, and eventually lands at the base of the cliff. Find the total time of flight and its speed just before hitting the ground.

37. A stone is dropped from the top of a tower. At the same instant, a second stone is thrown vertically upward from the ground with a velocity of 25 m/s. If the two stones meet at a height of 20 m above the ground, find the possible height(s) of the tower.

38. A lift is moving upward with a uniform velocity of 10 m/s. A bolt drops from the ceiling of the lift, 3.2 m above the lift floor. Find the time taken by the bolt to hit the floor of the lift, and explain why the lift's own velocity does not affect this time.

39. Two balls are thrown vertically upward from the ground, each with a speed of 25 m/s, the second one thrown 1 s after the first. Find the height above the ground at which the two balls meet, and how long after the first throw this happens.

40. A parachutist free-falls for 5 s after jumping, then opens her parachute, which produces a uniform deceleration of 2.5 m/s^2 until she lands with a speed of 5 m/s. Find (a) her speed at the instant the parachute opens, and (b) the total height of the jump.

Motion Under Gravity — Answer Key

1. $t = 3$ s, $v = 30$ m/s

2. $t = 4$ s, $v = 40$ m/s

3. $t = 5$ s

4. 20 m; 80 m

5. $t = 2$ s

6. $t = 3$ s

7. $h = 100$ m

8. $h = 180$ m, $t = 6$ s

9. $h = 245$ m, $v = 70$ m/s

10. $t = 2$ s, $h = 28$ m

11. depth = 20 m

12. $t = 2$ s, $H = 20$ m

13. $H = 45$ m, $T = 6$ s

14. $u = 20$ m/s, $t = 2$ s

15. $u = 40$ m/s, $H = 80$ m

16. $v = -5$ m/s (moving down), $s = 3.75$ m

17. $v = 10$ m/s (still rising — time to top is 4 s)

18. $u = 30$ m/s, $H = 45$ m

19. $t = 1$ s and $t = 5$ s

20. $v = \pm 10$ m/s (10 m/s up, 10 m/s down)

21. $s = 30$ m, $v = 5$ m/s (still rising)

22. $v = -15$ m/s, i.e. 15 m/s downward

23. $s(1\text{ s}) = 10$ m, $s(2\text{ s}) = 10$ m — equal because 1 s and 2 s are symmetric about $t_{top} = 1.5$ s

24. $t = 5$ s, $h = 125$ m

25. 1 : 3 : 5

26. $t = 2.5$ s, $h = 31.25$ m

27. s(1st s) = 20 m upward; s(4th s) = 10 m, but downward (ball crosses the top at $t = 2.5$ s)

28. $t = 1$ s and $t = 3$ s; interval = 2 s

29. new speed = 2v; new time = 2t (since $v \propto \sqrt{H}$ and $t \propto \sqrt{H}$)

30. new height = 4H; new time of flight = 2T (since $H \propto u^2$ and $T \propto u$)

31. $v_1 : v_2 = 4 : 3$

32. velocity entering window = 9 m/s

33. $t = 2.5$ s; height = 68.75 m above the ground

34. separation = 80 m; grows linearly with time ($\propto t$) as both balls fall

35. $t = 3$ s, landing speed = 25 m/s

36. $T = 4$ s, speed on landing = 25 m/s

37. $H = 25$ m (meeting at $t = 1$ s) or $H = 100$ m (meeting at $t = 4$ s) — both are valid

38. $t = 0.8$ s; independent of the lift's velocity since the lift moves at constant (non-accelerating) speed, so the relative acceleration of the bolt is still g

39. meet at $t = 3$ s after the first throw; height = 30 m

40. (a) 50 m/s (b) total height ≈ 620 m (125 m free-fall + 495 m under parachute)